S.S. DHARANEE DHARAN

DME, BE (Mechatronics) | Mechatronics Design Engineer | Exp: 4.5 Years

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Location: Namakkal, Tamil Nadu

ABOUT ME

A results-driven Mechatronics Design Engineer specializing in robotics, automation, and IIoT-based systems. Proven experience in the design and development of real-time engineering solutions, including electric vehicles (EVs), collaborative robots, and Industry 4.0 modular manufacturing systems. Dedicated to delivering innovative, efficient, and scalable automation and design solutions for industrial applications.

EDUCATION

- B.E. Mechatronics:
  - Bannari Amman Institute of Technology (2020 – 2023) – CGPA: 8
- Diploma in Mechanical Engineering:
  - Kongunadu Polytechnic College (2017 – 2020) – 85%

PROFESSIONAL EXPERIENCE

Mechatronics Design Engineer:

- Janatics India Pvt Ltd (Sep 2023 – Present)
  - Designed Industry 4.0 based mechatronics systems
  - Developed robotics systems and cobot trainer kits
  - Integrated PLC, HMI, pneumatics and IIoT
  - Built modular automation systems for real-time industrial workflows

INTERNSHIP EXPERIENCE

- TV Sundram Iyengar & Sons Pvt Ltd
  – Engine & ECU Diagnostics (2020 – 2021)
- Rinex Technology Pvt Ltd
  – Inline Sales Executive (2023 – 2023)

TECHNICAL SKILLS

- SolidWorks,
- Automation Studio,
- AutoCAD,
- Siemens NX,
- Windchill,
- SolidWorks Composer
- GD&T / Standards,

PROJECTS

Modular Manufacturing System (MMS) with IIoT Integration

- Designed and developed smart manufacturing modules using sensors, PLC, and cloud monitoring
- Tools Used: SolidWorks, AutoCAD, Automation Studio.

Autonomous Mobile Robot (AMR) with Cobot Integration

- Designed and developed AMR integrated with a collaborative robot for smart factory automation
- Tools Used: SolidWorks, PLC, HMI, Python, ROS

EV Agriculture Rotavator

- Designed and developed an electric rotavator for agricultural applications
- Tools Used: SolidWorks, AutoCAD, Automation Studio, Tinkercad, Ansys

Human–Cobot Chess Playing Station

- Designed and developed a Human–Cobot chess-playing system
- Tools Used: SolidWorks, PLC, HMI, Python, ROS

Cobot Trainer Kit Development

- Built a modular cobot training system for academic and industrial use
- Focused on safety, modularity, and real-time collaborative automation
- Tools Used: SolidWorks, AutoCAD, Automation Studio, Robodk.

Cobot Trainer Kit with Automatic Tool Changer (ATC)

- Designed and developed an advanced cobot trainer kit with automatic tool changing
- Tools Used: SolidWorks, AutoCAD, Automation Studio, Robodk.

ACADEMIC PROJECT

EV Car Design & Development

- Designed, built, and tested a fully functional electric vehicle prototype
- Won **2nd Prize** in a national-level innovation competition
- Contributed to powertrain integration, chassis design, and BMS development
- Performed real-time validation and on-track testing

HOBBIES

- Reading technical books
- Mechanical restoration of machines

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge.